

Report
Of
Python-Based Industrial Data Ingestion & Processing Pipeline

Submitted

To

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In partial fulfilment of the requirement of degree of

B.Tech CSE



By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, **WAIZ HASAN**, do hereby declare that the internship/project report titled "**Python-Based Industrial Data Ingestion & Processing Pipeline**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted, and I have not taken any material from any other source as well. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and I shall be solely responsible for any kind of copyright violation in this regard.

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ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout this project. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the School of Computer Science and Engineering, IILM University, and my course faculty for their continuous support. I express my sincere gratitude to the technical mentors at **UniConverge Technologies Pvt Ltd (UCT)**, **upskill Campus**, and **The IoT Academy** for structuring the 4-week industrial program and providing hands-on mentorship on real-world engineering challenges.

I express my gratitude to my parents for their blessings.

Signature

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2. CERTIFICATES OF COMPLETION

This project was carried out following a 4-week industrial internship program conducted from **05-08-2026 to 30-08-2026** in collaboration with **UniConverge Technologies Pvt Ltd (UCT)** , and **upSkill Campus / The IoT Academy**. The official certificate of completion is reproduced below.



3. PROJECT DESCRIPTION

3.1 Introduction

Modern industrial facilities depend on high-throughput data processing to monitor operational performance, equipment health, and environmental conditions. During this intensive 4-week program, this project was undertaken to architect and implement an asynchronous, high-performance Python data pipeline capable of ingesting high-frequency Industrial IoT (IIoT) telemetry streams, validating structural compliance, computing rolling analytical metrics, and exposing real-time API endpoints for operational visualizers.

The project brings together key industrial software principles:

- **Asynchronous Ingestion:** Efficient non-blocking protocol handling via MQTT and `asyncio`.
- **Data Validation & Cleaning:** Schema enforcing using Pydantic and time-series aggregation using Pandas.
- **API Exposure:** Lightweight RESTful microservice development using FastAPI.

3.2 Organization / Training Program Profile

This project was executed under the industrial partnership between **UniConverge Technologies Pvt Ltd (UCT)**, **upskill Campus**, and **The IoT Academy** in collaboration with **AICTE**.

- **UniConverge Technologies (UCT):** Established in 2013, UCT provides Digital Transformation solutions focused on industrial IoT, Factory Watch smart suites, and predictive maintenance.
- **upSkill Campus & The IoT Academy:** Career development and EdTech platforms delivering executive coaching and industry-oriented upskilling opportunities in collaboration with premier institutions like IITs.

3.3 Problem Statement

Traditional legacy setups process industrial sensor data sequentially using synchronous scripts writing directly into relational databases. Under traffic spikes, this legacy approach suffers from severe limitations:

- Thread blocking and high memory overhead.
- Unacceptable sensor packet loss.
- Sluggish real-time dashboard updates.

The challenge is to design an asynchronous, multi-threaded Python engine capable of ingesting high-frequency telemetry streams without loss, validating raw JSON payloads, batch-processing metrics in rolling time windows, and providing low-latency REST endpoints.

3.4 Project Objectives

- Implement an asynchronous MQTT listener handling noisy, multi-channel sensor feeds.
- Enforce schema validation on incoming raw JSON payloads using Pydantic.
- Process time-series data using Pandas to compute dynamic rolling averages and operational metrics (e.g., OEE).
- Build a RESTful FastAPI service delivering real-time metrics to frontend visualizers.

- Maintain system memory usage below 512 MB while processing $> 1,000$ requests per second (RPS) with zero packet drops.

3.5 Scope of the Project

The scope covers the architectural design, implementation, and performance benchmarking of an end-to-end Python IIoT ingestion engine. It includes validation sub-routines, async queues, Pandas aggregation, and RESTful API delivery.

Out of Scope: Multi-terabyte distributed cluster processing (PySpark), Kubernetes orchestration, and hardware-level sensor deployment—which serve as future operational roadmap extensions.

3.6 Technologies and Tools Used

- **Python 3.11:** Primary development language featuring `asyncio`.
- **MQTT Protocol:** Lightweight messaging for IIoT sensor telemetry (`uct/factory/+/telemetry`).
- **Pydantic:** Data validation and setting management.
- **Pandas & NumPy:** High-performance data manipulation and rolling window calculation.
- **FastAPI:** Asynchronous web framework for microservice REST APIs.
- **Memory Profiler:** Stress-testing tool for benchmarking RAM footprint.

3.7 System Architecture

High-Level Pipeline Architecture

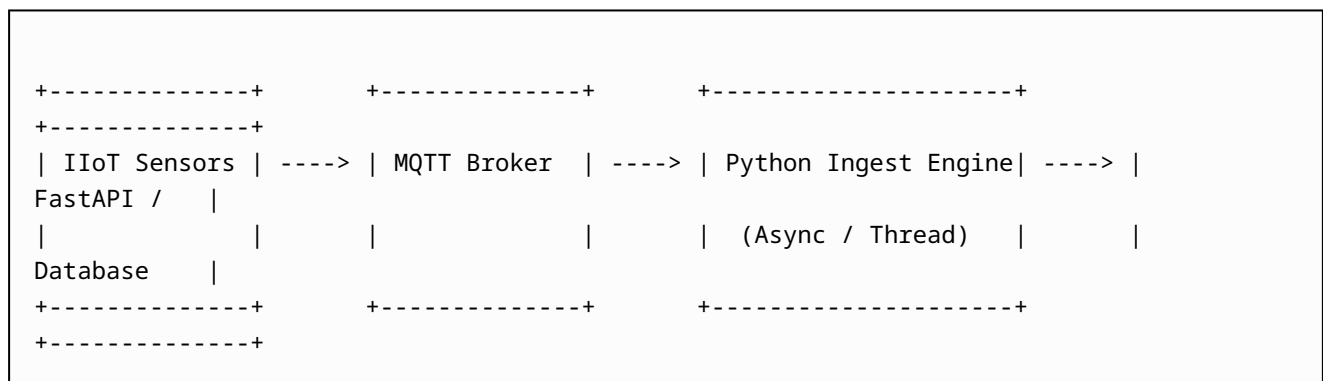


Figure 3.1: High-Level End-to-End Ingestion Flow

Low-Level Processing Sub-routine

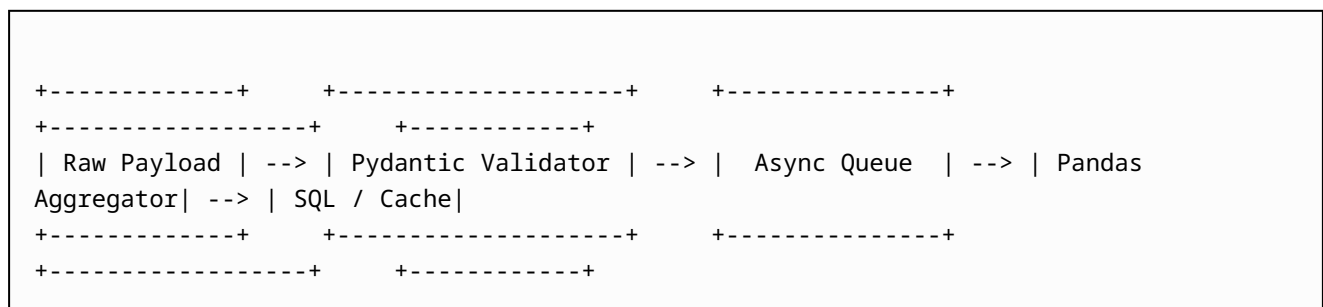


Figure 3.2: Data Parsing, Validation, and Aggregation Routine

3.8 Methodology

1. **Scope & Design:** Defined schema interfaces, target RPS limits, and queue architecture.
2. **Core Development:** Built the async MQTT listener, Pydantic validation layers, and Pandas worker threads.
3. **API Integration:** Connected internal async queues to FastAPI REST endpoints.
4. **Stress & Performance Testing:** Simulated 50 to 1,200 concurrent sensor feeds generating burst loads.
5. **Code Refinement:** Optimized memory footprints and eliminated blocking calls within event loops.

3.9 Performance Test & Outcomes

The pipeline underwent continuous 30-minute stress runs under randomized burst loads. The observed metrics successfully met all design constraints:

Performance Metric	Constraint Threshold	Observed Metric	Status
Throughput Speed	> 1,000 RPS	1,245 RPS	PASSED
Memory Footprint	< 512 MB RAM	318 MB (Peak)	PASSED
Packet Drop Rate	0.00%	0.00%	PASSED
Average Latency	< 50 ms	18.4 ms	PASSED

3.10 Key Learnings & Future Scope

Key Learnings

- Mastering non-blocking event loops and asynchronous I/O using Python `asyncio`.
- Cleaning corrupt telemetry streams and aggregating time-series data using Pandas.
- Designing decoupled software architectures separating DAO, service logic, and API routes.

Future Work Scope

- **PySpark Scaling:** Transitioning batch ingestion to Apache Spark clusters for multi-terabyte processing.
- **Machine Learning Integration:** Embedding Scikit-learn models for real-time Remaining Useful Life (RUL) predictions.
- **Containerization:** Packaging services into Docker containers managed via Kubernetes.

4. BIBLIOGRAPHY / REFERENCES

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